Bridging the Gap in Infrastructure Modeling & Visualization

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I. Introduction

Within the United States, there are more than 617,000 bridges each of which is paramount to ensuring the safe and reliable transit of goods, services, and individuals throughout the country. Despite their critical function, a significant proportion of the nation's bridges are in a state of disrepair, creating an issue of substantial public concern. However, this issue is not often recognized in part due to the lack of non-proprietary solutions for transparent exploration and visualization of the topic. This gap in existing technology creates the need for an open-source user-friendly interface for both experts and non-experts alike to better visualize, understand, and explore both existing national bridge data [1] and deterioration modeling approaches. The problem and how we intend to solve it is thus two-fold, composed of enabling a unified modeling and visualization framework for the United States bridge inventory systems that improve upon present solutions.

II. Modeling

From a modeling perspective, three general categories broadly encompass the present body of academic work in bridge condition modeling: deterministic models, probabilistic models, and deep neural networks models. Deterministic models incorporate a set of defined relationships among features like time, age, region, etc. derived from human feature engineering and selection that would be used to model a given bridge component's need for replacement or repair based upon thresholds determined by engineers [2]. These initial approaches proved far too simplistic and inefficient for the task, leading to over replacement and poor resource allocation of components and labor. Gradually, their complexity increased through the adoption of linear regression models [3], polynomial regression models [4], and ensemble methods such as Random Forests [5]. However, their utility is limited by their rigid structure, susceptibility to multicollinearity, and reliance upon non-scaleable and often biased human feature engineering. Efforts have been made to remedy these shortcomings through variable selection methods such as LASSO and Step-wise Selection [6] as well as dimensionality reduction methods such as tensor decomposition [7]. Although effective in practice, this class of models is unable to quantify uncertainly with respect to inputs or predictions. This quality, which is key to risk assessment, is the primary reason for the creation of the probabilistic class of models. Initial probabilistic models were based upon failure and reliability distributions [8] parameterized by known sources of failure over a bridge's lifetime. However, due to the infrequent nature of bridge collapses and the inability of such an approach to dynamically adjust to temporally realized conditions, these approaches were replaced with more flexible stochastic process models. Presently the most utilized among these models are Markov chains [9] and as an extension of it, Bayesian Decision Networks [10] both of which permit the modeling of a bridge concerning its present known condition, a major improvement upon the static probabilistic models. However, the Markovian approach does not solve the issue of time dependence entirely as the process itself is memory-less, only the current state influences the next system state. This assumption directly contradicts the known dynamics of deep temporal dependencies throughout the deterioration process [11]. Furthermore, similar to early generation models there is still significant subjective feature engineering and distributional approximation taking place in the construction of the transition probability matrices of these decision networks and Markov processes [12]. The most recent solution of AI models directly addresses this issue as well as the necessity of expert feature engineering which plagues both deterministic and probabilistic models. This is most commonly achieved through the utilization of deep artificial neural networks. These computational models utilize several processing layers to learn and create a hierarchy of abstraction to best represent the training set [13]. A wide variety of neural network architectures have been proposed from dense models [14] to convolutional networks [15] that more effectively engineer features through up-sampling of the feature space. Lesser known, but still relevant alternative hybrid AI methods include clustering approaches through KNN [16] and K-means [17] [5] as a feature engineering or prepossessing step to more effectively apply subsequent probabilistic modeling approaches. Furthermore, AI methods are far less transparent especially to non-expert users compared to the prior two classes of models, and still do not provide a complete solution to the temporal aspect of the problem. Despite the substantial body of work that composes present modeling approaches for bridge deterioration and forecasting, there is still significant room for improvement. The most prevalent short-coming which can be attributed to all contemporary modeling approaches is the lack of data integration across some of the most crucial known causes of bridge deterioration such as ambient temperature [18], climate [19], traffic, and accident data [20], and utilizing only a fraction of the national data available. Incorporating and unifying all of the existing national data and replacing traffic and climate estimates with the real observed data sources from NOAA [21] and the FHA [22] will lead to both more accurate and comprehensive models. In addition to these data augmentations, we plan to leverage all the insights of past approaches and model classes to deliver a hybrid model capable of improving the state of the art through improved feature selection and treatment of the data's sequential nature. We aim to use an iterative design approach to prototype and test several models, which will ultimately allow us to avoid any pitfalls of a predefined model type/structure. The only real risk faced by our project is the potential for over-fitting which can be dangerous especially when the public trusts and depends on it for information. However, these risks are present in any model and can be reduced through careful design and cross-validation which will also effectively test the model's success.

III. **Visualization**

From a visualization perspective, the single freely available solution that exists today is the "InfoBridge" web portal [23]. The portal itself has data stored in a format not tailored to visualization design principles. Column names make data both inaccessible and confusing to expert and non-expert users alike and current geographic mapping lacks any interactivity or additional meaning beyond general location making the visualization uninformative. Performance and projection charts are visually unappealing and uninformative given the layout, size, and design of their associated sections thus limiting data exploration potential. In its present state, the platform is extremely inaccessible to consumers, and we plan to improve it through a first-principles redesign of the user interface with a focus on data interactivity. In contrast to current solutions, we will create a unified framework for both modeling and visualization. Additionally, we will leverage Google street-view API [24] to provide a more meaningful and localized depiction of each of the nation's 617,000 bridges with drill-downs for discovering additional bridge-specific information. Putting a real view of each data point makes the visualization both more substantive and interactive. Verifiable improvement of our redesigned interface will be evaluated with user studies and via visualization effectiveness metrics [25].

IV. **Development**

From a development perspective, we plan for the project to take around seven weeks within our six-person team. Associated costs of the project are limited to AWS credits with no other expenses foreseen. Development will be broken into three sub-teams corresponding to data, modeling, and visualization. Each will focus on the implementation of their respective features discussed in the body of the report. The data team, composed of Ben and Jacob, will be responsible for cleaning and integrating our various data sources, providing sample data sets for development within the modeling and visualization teams, and creating and maintaining the necessary SQL databases needed for data storage and retrieval. The modeling team, composed of Evan and Mernegar, will be responsible for developing, training, and cross-validating models of bridge deterioration. They will also be responsible for back-end optimization efforts to ensure reliable and fast prediction is feasible within the application. The visualization team, composed of Brian and Travis, will be responsible for creating a new user interface which will include capabilities such as national choropleth of bridge health, statewide reports, and comparisons, interactive tool-tips, and visualization of street-view images of specific bridges. Decisions on specific technologies utilized will be made on a sub-team basis, but tentatively comprise MySQL for database storage, python and flask for the web application framework, and Bokeh for interactive visualization. Progress will be monitored throughout via "midterm" and "final exam" deliverable checkpoints corresponding to the first and second months of the project schedule. The midterm checkpoint on week 3 will be composed of each team presenting their isolated features discussed above, and the remainder of the project time will involve the integration of the three teams' work thus far. The final exam checkpoint on week 6 will be a test of complete project integration and demonstration followed by subsequent work on final presentation deliverables.

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